

SUMANTH KUMAR SIVADI

Machine Learning Engineer | Data & ML Platform Engineering

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[LinkedIn](#) | [\[GitHub URL\]](#) | [\[Portfolio URL\]](#)

Authorized to work in the United States

PROFESSIONAL SUMMARY:

AI Engineer with 4+ years of experience delivering end-to-end AI/ML solutions across healthcare and financial services. Translated business and clinical requirements into predictive models, fraud and credit-risk solutions, clinical decision-support systems, and patient outcome analytics. Built and optimized healthcare and financial data pipelines, engineered features, trained and validated ML, deep learning, NLP, and transformer models, and implemented RAG and AI-agent workflows for unstructured data and knowledge retrieval. Deployed models into production, established MLOps and monitoring practices, and built solutions that improved processing efficiency, model accuracy, and operational decision-making.

TECHNICAL SKILLS:

Languages:	Python, SQL, R, Scala, Bash/Shell Scripting
AI/ML:	Machine Learning, Deep Learning, NLP, Reinforcement Learning, Transfer Learning, Classification, Regression, Clustering, Recommendation Systems, Time Series Forecasting, Anomaly Detection, Feature Selection, Hyperparameter Tuning
ML Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, OpenCV, spaCy, NLTK
GenAI/LLMs:	LLMs, RAG, Fine-Tuning, Prompt Engineering, AI Agents, Agentic AI, Multi-Agent Systems, OpenAI API, Llama Models, LangChain, LlamaIndex, LangGraph, Embeddings, Semantic Search
Data Engineering:	Pandas, NumPy, PySpark, Apache Spark, Apache Kafka, Apache Airflow, ETL/ELT, Data Modeling, Data Cleaning, Data Validation, Feature Engineering, Power BI, Tableau
Cloud:	AWS, Microsoft Azure, Amazon SageMaker, Azure Machine Learning, Azure OpenAI, Azure Databricks, Azure Data Factory, Lambda, EC2, S3, Glue, Redshift
MLOps/Deployment:	MLflow, Kubeflow, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, GitLab, Model Monitoring, Experiment Tracking, Automated Retraining, FastAPI, Flask, REST APIs, Microservices
Databases:	PostgreSQL, MySQL, SQL Server, Oracle, MongoDB, Snowflake, Redis, Elasticsearch, Pinecone, ChromaDB, FAISS
Development/Agile:	Git, GitHub, VS Code, Linux/Unix, Postman, Agile, Scrum, Jira, Confluence, A/B Testing
Security/Governance:	Data Privacy, Model Security, Bias Detection, Model Explainability, RBAC, HIPAA, GDPR, SOC 2, Secure AI Deployment, AI Governance

PROFESSIONAL EXPERIENCE:

Johnson & Johnson | AI Engineer | USA | Aug 2024 – Present

- Translated clinical and business requirements into AI/ML use cases for clinical decision support, disease-risk prediction, and patient outcome analysis, collaborating with healthcare and business stakeholders in Agile/Scrum teams.
- Assessed and integrated structured and unstructured healthcare data, addressing data-quality issues and preparing reliable datasets for clinical analytics and model development.
- Re-engineered large-scale healthcare data pipelines using Python, PySpark, and SQL, reducing data processing time by 35% and improving data availability for downstream analytics.
- Performed exploratory analysis, feature engineering, and model validation to prepare clinical datasets for predictive modeling and improve the reliability of healthcare AI solutions.
- Built and evaluated machine learning and deep learning models for disease-risk prediction, patient outcomes, and clinical intelligence, improving model response accuracy by 25%.
- Applied NLP and transformer-based models to process unstructured clinical information, extracting relevant insights from healthcare documents and supporting clinical decision-making.
- Engineered RAG-based information retrieval workflows using LLMs and enterprise knowledge sources to deliver context-aware responses and improve access to relevant clinical information.
- Implemented AI-agent workflows to automate document processing, information retrieval, and repetitive clinical support activities, reducing manual intervention across healthcare workflows.
- Deployed AI solutions as scalable services and integrated them into production workflows, supporting automated model delivery and lifecycle management across the AI environment.

- Built dashboards for real-time model performance and data-quality monitoring, enabling proactive drift detection and faster issue resolution while supporting explainability, bias monitoring, and responsible AI governance.

Infosys | Machine Learning Engineer | India | Jan 2021 – Aug 2023

- Translated business requirements into ML solutions for fraud prevention, credit risk assessment, customer intelligence, and financial forecasting, working with risk and engineering teams to define measurable outcomes.
- Integrated transaction, customer, account, and payment data to create reliable datasets for analysis, reducing data processing time by 47% through optimized Python, PySpark, and SQL pipelines.
- Analyzed transaction behavior and historical outcomes to identify fraud patterns, risk indicators, anomalies, and customer trends, guiding the selection of relevant predictive variables.
- Engineered behavioral and transaction-based features that captured spending patterns, account activity, and unusual transactions, improving fraud detection accuracy by 18%.
- Built and validated predictive models for fraud, credit risk, and forecasting, selecting approaches based on accuracy, reliability, and business impact rather than model complexity.
- Designed real-time and batch scoring workflows that evaluated transactions and customer activity, helping risk teams identify potentially fraudulent activity and prioritize higher-risk cases.
- Put approved models into production on AWS, establishing secure prediction services that supported consistent, low-latency risk and fraud decisions.
- Established MLOps practices for model versioning, experiment tracking, and automated deployments, making model releases more consistent, traceable, and easier to maintain.
- Monitored production models for performance changes and data drift, triggering retraining when necessary and reducing prediction latency by 30% while maintaining model reliability.
- Created Tableau dashboards and SQL reporting for fraud trends, risk metrics, and model performance, giving business stakeholders clear visibility into outcomes and supporting data-driven decision-making.

ACADEMIC PROJECTS:

VisionStack: Business Growth Analytics Platform - Python, PySpark, SQL, Databricks, Apache Spark, Git

- Built a scalable big data pipeline with PySpark and Databricks, processing millions of customer and product records from CSV, JSON, and Parquet sources into a unified analytics model.
- Designed SQL ETL workflows with deduplication, normalization, and validation, generating KPIs for quarterly sales trends, product performance, and customer segmentation

EDUCATION:

Master of Science, Applied Computer Science - Northwest Missouri State University, Maryville, **MO** | GPA 3.9/4.0

B.Tech, Electronics & Communication Engineering - Audisankara College of Engineering and Technology, **India**